EXPERIMENT 7: SLIDING WINDOW PROTOCOL

AIM

To implement flow control at the data-link layer using the Sliding Window Protocol and simulate the flow of frames from one node to another.

THEORY

The Sliding Window Protocol allows a sender to transmit multiple frames before receiving acknowledgements. The window size determines the number of outstanding frames. ACK confirms successful reception, while NACK indicates that a frame or sequence needs retransmission.

A. SENDER PROGRAM (sender.py)

```
# sender.py
import time

window = int(input("Enter window size: "))
message = input("Enter message: ")
frames = [{"Frame": i, "DATA": ch} for i, ch in enumerate(message)]

with open("Sender_Buffer", "w") as f:
    for start in range(0, len(frames), window):
        batch = frames[start:start + window]
        for frame in batch:
            f.write(f'{frame["Frame"]},{frame["DATA"]}\\n')
            print("Sent:", frame)
        time.sleep(1)  # simulate transmission delay

        try:
            with open("Receiver_Buffer") as f:
                ack = f.read().strip()
        except FileNotFoundError:
            ack = "NACK"

        expected_ack = str(batch[-1]["Frame"] + 1)
        if ack == expected_ack:
            print("ACK received:", ack)
        else:
            print("NACK/incorrect ACK; resending window")
            with open("Sender_Buffer", "w") as f:
                for frame in batch:
                    f.write(f'{frame["Frame"]},{frame["DATA"]}\\n')
            time.sleep(1)

print("Transmission completed.")
```

B. RECEIVER PROGRAM (receiver.py)

```
# receiver.py
with open("Sender_Buffer") as f:
    lines = [line.strip() for line in f if line.strip()]

expected = 0
received = []

for line in lines:
    frame_no, data = line.split(",", 1)
    frame_no = int(frame_no)
    if frame_no == expected:
        received.append(data)
        expected += 1
        status = f"ACK,{expected}"
    else:
        status = f"NACK,{expected}"

    with open("Receiver_Buffer", "w") as f:
        f.write(status.split(",")[1])
    print(f"Frame {frame_no}: {status}")

print("Received message:", "".join(received))
```

SAMPLE INPUT / OUTPUT

```
Enter window size: 3
Enter message: HELLO
Sent: {'Frame': 0, 'DATA': 'H'}
Sent: {'Frame': 1, 'DATA': 'E'}
Sent: {'Frame': 2, 'DATA': 'L'}
ACK received: 3
Sent: {'Frame': 3, 'DATA': 'L'}
Sent: {'Frame': 4, 'DATA': 'O'}
ACK received: 5
Transmission completed.

Receiver:
Frame 0: ACK,1
Frame 1: ACK,2
Frame 2: ACK,3
Frame 3: ACK,4
Frame 4: ACK,5
Received message: HELLO
```

RESULT

The Sliding Window Protocol was implemented to simulate frame transmission, acknowledgement handling, and flow control between sender and receiver.